PROJECT REPORT: KEFFI AI

CLINICAL DIGITAL THERAPEUTICS & REAL-TIME BIOFEEDBACK-SYNCED CBT PLATFORM

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NAAN MUDHALVAN - NIRAL THIRUVIZHA 3.0 (2025 - 2026)

FINAL PROJECT REPORT

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PROJECT TITLE : KEFFI AI - CLINICAL DIGITAL THERAPEUTICS & REAL-TIME BIOFEEDBACK-

SYNCED COGNITIVE BEHAVIORAL THERAPY (CBT) PLATFORM

MAJOR AREA : Artificial Intelligence in Mental Healthcare

(AI-Based Clinical Decision & Therapeutic Support System)

PROBLEM STATEMENT: How might we utilize AI chatbots and machine learning to address

the challenges of incomplete alleviation of depression symptoms,

attrition, and loss of follow-up in mental health treatment.

INSTITUTION : UNIVERSITY COLLEGE OF ENGINEERING, BIT CAMPUS, ANNA UNIVERSITY

TIRUCHIRAPPALLI - 620024

DEPARTMENT : DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (CSE)

GUIDED BY : Dr. SIVANESH, B.Tech., M.Tech., Ph.D., Assistant Professor / CSE

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DATE OF REVIEW : 04.08.2026 to 07.08.2026

HOST INSTITUTION : ANNA UNIVERSITY, BIT CAMPUS, TIRUCHIRAPPALLI

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1. ABSTRACT

Keffi AI is a Clinical Digital Therapeutics Platform engineered to overcome the critical limitations of standard AI wellness chatbots, which typically suffer from a lack of long-term memory, absence of clinical safety protocols, and inability to track physiological stress.

The platform incorporates:

• A BERT-based fine-tuned emotion classifier capable of recognizing 96 clinically granular emotional states, far surpassing binary positive/negative sentiment analysis.

• A Vector-based Memory Engine (Pinecone + embeddings) that builds a continuous psychological profile of the user across multi-session interactions.

• A real-time Mental Health Quotient (MHQ) metric and predictive attrition algorithm that identifies early emotional decline and engagement drop-offs.

• An automation layer (n8n) that triggers real-time clinical interventions, smart reminders, acoustic sanctuary audio playback, and emergency escalation lines.

• A Hardware Biofeedback Interface syncing live heart-rate PPG sensor data to dynamically adjust CBT techniques during panic episodes.

Keffi AI bridges the gap between self-help conversational tools and professional psychiatric care, providing a safe, accessible, and evidence-based 24/7 therapeutic companion.

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2. PROBLEM STATEMENT & THREE INTERCONNECTED CLINICAL CHALLENGES

2.1 Incomplete Alleviation of Depression Symptoms

Standard psychiatric treatments (pharmacotherapy and weekly psychotherapy) work for many patients but fail to achieve full remission in 50–60% of cases following first-line treatment. This occurs because:

• Therapy is often episodic and one-size-fits-all.

• Symptom tracking between clinic appointments is inconsistent or non-existent.

• Patients suffer in isolation between scheduled visits, allowing symptoms to escalate undetected.

2.2 Attrition (Patient Dropout Rate)

A significant proportion of patients abandon mental health treatment prematurely due to:

• Social stigma surrounding traditional therapy visits.

• High cost and geographic accessibility barriers.

• Lack of visible, quantified progress during early treatment phases.

• Feeling disconnected from healthcare providers seen infrequently.

2.3 Loss to Follow-Up

Depression is an inherently recurring condition. Patients who discontinue care frequently experience undetected relapses because clinicians are overwhelmed and unable to manually track hundreds of patients continuously.

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3. PROPOSED SOLUTION - KEFFI AI ARCHITECTURE

Keffi AI operates as a clinical companion using 7 scientifically validated therapeutic AI response strategies:

1. Validation & Empathy (Person-Centered Therapy): Accepts user feelings without judgment to establish trust (proven to account for 30% of therapeutic outcome).

2. Metaphor & Storytelling (ACT): Uses cognitive defusion techniques to reduce overthinking.

3. Grounding & Somatic Techniques (Polyvagal & MBSR): Delivers 4-7-8 breathing exercises and acoustic frequency audio (proven to reduce panic by 45–50%).

4. Cognitive Reframing (CBT): Identifies cognitive distortions and reframes negative thinking patterns.

5. Reflective Questioning (Motivational Interviewing): Prompts self-awareness through targeted open-ended questions.

6. Guided Options & Quick Replies: Reduces cognitive effort and decision fatigue during acute distress.

7. Crisis Intervention & SOS Protocol: Detects suicidal or high-risk telemetry signals, deploying WHO-compliant emergency hotline interfaces and human therapist scheduling popups.

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4. SCIENTIFIC RESEARCH & CLINICAL VALIDATION

• Hofmann et al. (2012): The Efficacy of Cognitive Behavioral Therapy: A Review of Meta-analyses. Demonstrates 50–75% reduction in depression/anxiety symptoms.

• Norcross & Wampold (2011): Evidence-Based Therapy Relationships. Confirms empathy and therapeutic alliance drive 30% of clinical success.

• Hayes et al. (2006): Acceptance and Commitment Therapy. Proves MBSR and ACT grounding techniques lower panic symptoms by 45–50%.

• Fitzpatrick et al. (2017): Delivering CBT to Young Adults Using Conversational Agents (Woebot Trial). Showed a 22% reduction in depression symptoms in just 2 weeks of automated conversational agent interaction.

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5. FINANCIAL UTILIZATION SUMMARY (■15,000.00)

S.No	Component / Vendor	Invoice No	Date	Amount (■)
1	Cloud GPU Server Instances for BERT Emotion Fine-Tuning	CNS/2026/0482	12-06-2026	4,500.00
2	Biofeedback Hardware PPG Sensor & ESP32 Microcontroller Kit	ETC/25-26/1104	05-06-2026	3,800.00
3	LLM API Tokens, Pinecone Vector Database & n8n Automation	WSS/2026/089	20-06-2026	3,200.00
4	Production Web Domain, SSL Certificate & Cloud Gateway	CHM/2026/0312	02-07-2026	1,500.00
5	Project Report Printing (3 Copies), Gold Embossing & Binding	HTP/2026/7841	25-07-2026	2,000.00
TOTAL				■15,000.00

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6. CONCLUSION & ACKNOWLEDGEMENTS

Keffi AI demonstrates how modern artificial intelligence, when grounded strictly in clinical literature and biofeedback integration, can expand mental healthcare access, prevent patient dropout, and support certified clinicians.

We express our deep gratitude to Tamil Nadu Skill Development Corporation (TNSDC), Naan Mudhalvan Niral Thiruvizha 3.0, Anna University, and our Faculty Guide Dr. Sivanesh for their guidance and financial support.